NephroTwin — System Architecture

Proof-of-concept architecture using synthetic data. The output is an illustrative index, not a clinically validated prediction.

1. Historical profile patient_static.csv Age, fictional CKD label, baseline eGFR, history	**2. Dynamic stream** wearable_timeseries.csv Hourly heart rate, steps, temperature, humidity, simulated hydration
↓	↓
3. Data ingestion pandas CSV loading timestamp sorting and input checks	**4. Virtual patient state** digital_twin.py transparent hand-set index + 24-hour scenario simulation
■	■
5. Clinician-facing demo UI app.py / Streamlit profile, time-series charts, scenario controls, forecast comparison	**6. Validation & governance** tests, documentation synthetic data label, limitations, future clinical validation plan

Data flow: generate_data.py → synthetic CSV files → app.py reads static + dynamic data → digital_twin.py calculates illustrative index and scenarios → Streamlit dashboard renders results. No real patient data or clinical decision support is used.